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Mathematical model and experimental validation of surface profile of a workpiece in round–oval–round pass sequence

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Abstract

A reliable analytical model that determines the surface profile of a workpiece (material) in round–oval–round pass sequence has been established. The surface profile of an outgoing workpiece is computed by using the linear interpolation of the radius of curvature of an incoming workpiece and that of roll groove to the roll axis direction.

The validity of the surface profile model has been examined by hot rod rolling experiment with the roll gap and the design of oval groove changed. The surface profile and cross sectional area of a workpiece predicted by the new model were in agreement with those obtained experimentally. © 2000 Published by Elsevier Science B.V.

Keywords: Rod rolling; Surface profile; Analytical model; Cross sectional area

1. Introduction

In continuous hot rod rolling process, billets are processed into rods with acceptable dimensional tolerance as they pass through the rolling stands, with the cross sectional shape being progressively altered. In the past, to predict the surface profile of workpiece at a pass (stand), a process designer has relied on an empirical equation based on tryout, which requires high cost and a lot of time when designing the process geometry defined by roll profiles and roll gap.

Since 1990, a number of numerical studies that simulate rod rolling have been presented on the basis of three-dimensional finite element analysis [1–4]. The FEA is very effective in calculating the surface profile and the cross sectional area of workpiece but requires at least several hours to run a program for a single pass since three-dimensional analysis is required in nature. To calculate the cross sectional area of a workpiece at each pass quickly, the development of an analytical model is absolutely required, which can also predict the surface profile of the workpiece.

Kemp [5] proposed a deformation model to calculate the surface profile of a workpiece in oval and round groove rolling, and compared the experimentally determined surface profile of the workpiece with those calculated by the model. Kemp, however, did not present the equations for the prediction of surface profile used in the model.

In this study, a new analytical model that predicts the surface profile of a workpiece in oval and round groove rolling has been proposed. The surface profile of a workpiece can be modeled when the maximum spread of it is known beforehand. Then, the surface profile of an outgoing workpiece is computed by using the linear interpolation of the radius of curvature of an incoming workpiece and that of roll groove to the roll axis direction. Emphasis is placed on the aspects that the new model presented is neither made up of any empirically based equations nor based on experimental data.

The surface profile of a deformed workpiece depends on the spread (or elongation) of the workpiece. This must be sensitive to the material, rolling speed, gap/stock geometry and particularly friction. The experiment/model in this study take two cases, changing groove shape and roll gap. The rolling temperature and friction conditions between materials and roll grooves were fixed because rolling temperature change (950–1150°C) did not play a significant role in the spread of workpiece [6].

The validity of the surface profile model presented has been examined by round–oval–round pass rolling experiment. To investigate the effect of groove shape

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Nomenclature

A_h	area fraction of workpiece above roll groove when the cross section of workpiece and roll groove are overlapped
A_S	area fraction of workpiece cut out by B_c at the inside roll groove when the cross section of workpiece and roll groove are overlapped
B_c	interval of two cross points to the roll axis direction when the cross section of workpiece and roll groove are overlapped
B_x	x-co-ordinate of cross point of surface profile and oval groove
B_y	y-co-ordinate of cross point of surface profile and oval groove
C_x	x-co-ordinate of cross point of workpiece and roll groove
C_y	y-co-ordinate of cross point of workpiece and roll groove
D_r	roll depth
D_x	distance between the center of round groove and that of surface profile
G	roll gap
H_i	maximum height of incoming workpiece
$\bar{H}_i$	equivalent height of incoming workpiece
$\bar{H}_o$	equivalent height of outgoing workpiece
H_p	pass height
R_1	radius of curvature of oval groove
R_f	radius of final surface profile of oval groove
R_g	radius of curvature of round groove
R_{max}	maximum radius of roll
R_{mean}	mean radius of roll
R_S	radius of surface profile of workpiece at a pass
W_i	maximum width of incoming workpiece
W_f	face width of oval groove
W_{max}	maximum spread of workpiece at a pass
Greek letters	
α	relief angle of round groove
γ	correction coefficient dependent on pass type

on the model, two different shapes of oval groove were designed, which have different radii of curvature, R_1. The surface profile of the workpieces measured at room temperature after each pass was compared with those calculated by the model.

2. Problem formulation

The formulation and solution method are based on the maximum spread model and the linear interpolation of the geometry of an incoming workpiece and roll grooves. The followings gives the procedure to compute the surface profile and the cross sectional area of the deformed workpiece.

2.1. Maximum spread model in oval–round (or round–oval) pass rolling

Shinokura and Takai [6] carried out an important experiment using a pilot hot rolling mill and developed a model that calculates the maximum spread of workpiece in round–oval (or oval–round) pass and square-diamond (or diamond-square) pass rolling. The idea behind their model was that the maximum spread of an outgoing workpiece could be computed from roll radius, maximum size of an incoming workpiece, the area fraction between the incoming workpiece and the geometry of roll groove. Figs. 1 and 2 illustrate the definition of area fraction and equivalent heights in round–oval and oval–round pass rolling, respectively. The maximum spread (W_{max}) was calculated as follows:

$$W_{max} = \gamma(1 + W_i)\frac{R_{mean}(\bar{H}_i - \bar{H}_o)}{W_i + 0.5H_i}\frac{A_h}{A_o} \quad (1)$$

where

$$\bar{H}_o = \frac{A_o - A_S - A_h}{B_c} \quad \text{and } \bar{H}_i = \frac{A_o - A_S}{B_c} \quad (2)$$

W_i and H_i are the maximum width and the maximum height of an incoming workpiece,respectively and γ is correction coefficient dependent on pass type. $\gamma = 0.83$ is used for both oval–round and round–oval pass rolling in the present study.

2.2. Surface profile and cross sectional area of workpiece in oval–round pass rolling

Fig. 3 shows an oval workpiece incoming and a round groove. D_r and G are, respectively, roll depth and roll gap, α is the relief angle designed for workpiece in roll groove to flow out smoothly in the case of overfilling. R_1 is the radius of curvature of the oval workpiece incoming, $R_g$the radius of round groove and is also assumed as a final surface profile of workpiece after rolling. In oval–round pass, one of the possible surface profiles of workpiece is illustrated in Fig. 4. R_S is the radius of the surface profile of workpiece at the roll throat and is modeled by the linear interpolation of R_1 and R_g when the maximum spread, W_{max}, is known, i.e.

$$R_S = R_1 W_t + R_g(1 - W_t) \quad (3)$$

where

$$W_t = \frac{2R_g - W_{max}}{2R_g - W_i} \quad (4)$$

W_t is a weighing function of R_1 and R_g. The requirements we placed on the choice of the weighing function stem from the need to ensure Eq. (3). When R_S is calculated, the distance between the center of round groove and that of the surface profile is obtained as

$$D_x = \tfrac{1}{2}(R_S - W_{max}) \quad (5)$$

The position of D_x is always along the x-axis.

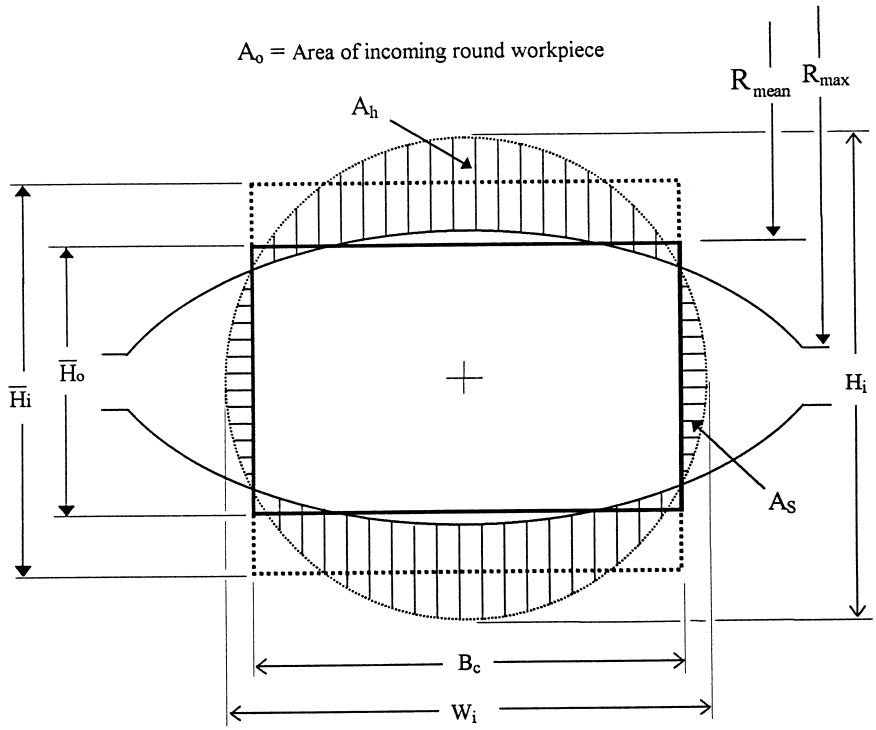


Fig. 1. Application of equivalent rectangle approximation into round–oval pass to calculate the effective height for workpiece, roll groove and roll diameter.

To confirm the validity of Eq. (3), boundary conditions of the proposed model are examined. If the incoming oval workpiece with the radius of curvature R_1 is not deformed at all, W_{max} (Fig. 4) is just W_i. Then, $W_t = 1.0$ is obtained from Eq. (4). Consequently, Eq. (3) yields that $R_S = R_1$, which makes sense. On the other hand, when the incoming oval workpiece is deformed and the maximum spread, W_{max}, reaches round groove diameter ($=2R_g$), Eq. (4) yields that $W_t = 0.0$. Then, Eq. (3) gives that $R_S = R_g$, which is right. It should be noted that Eqs. (3)–(5) are valid under the condition that the maximum spread of the outgoing workpiece is not greater than round groove diameter, and $(D_f + G/2)$ is equal to R_g. If $(D_f + G/2)$ is not equal to R_g, the round roll

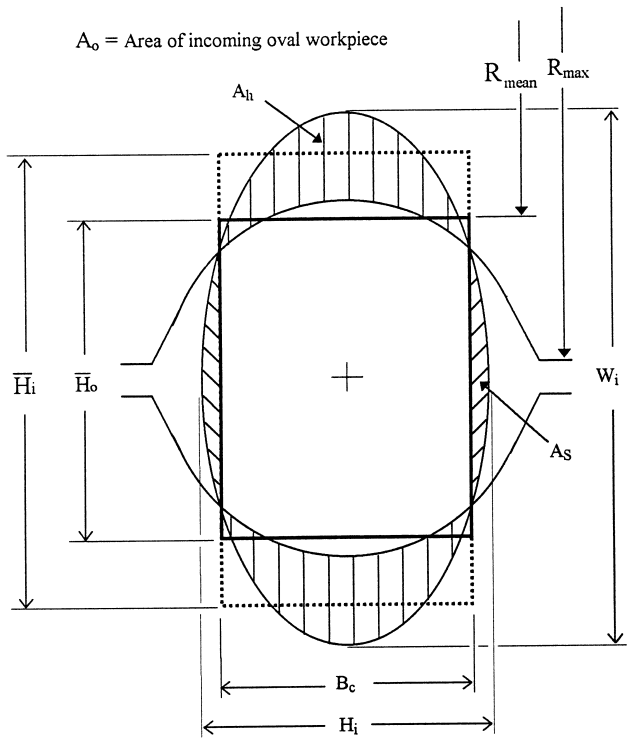


Fig. 2. Application of equivalent rectangle approximation into oval–round pass to calculate the effective height for workpiece, roll groove and roll diameter.

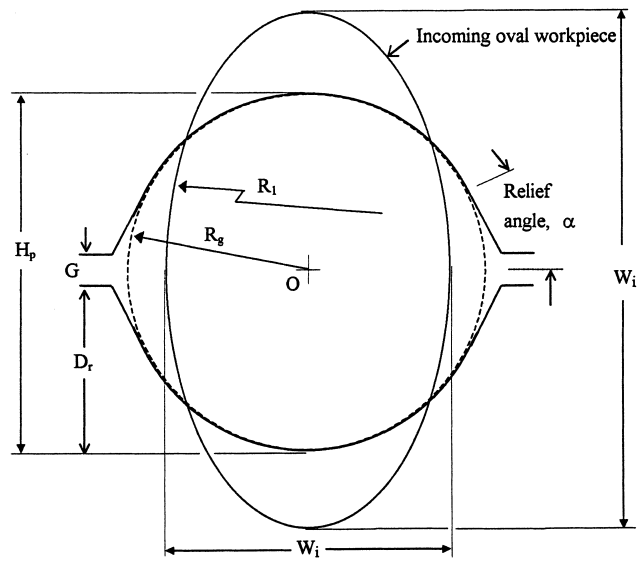


Fig. 3. An oval workpiece incoming and a round groove.

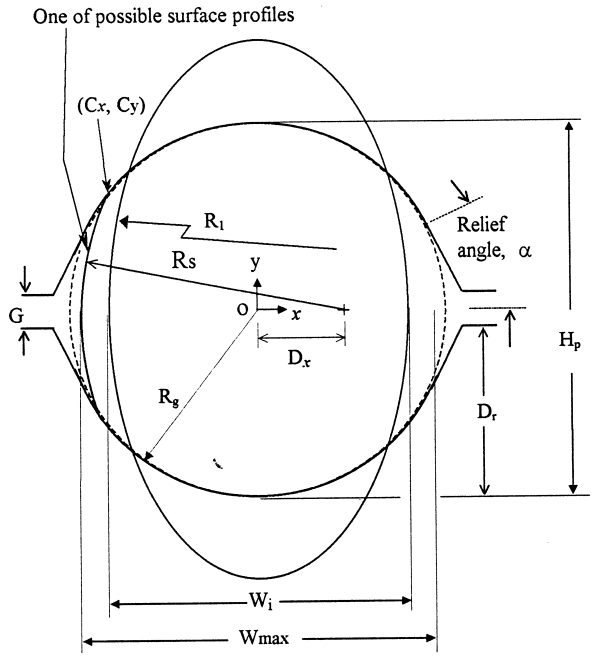


Fig. 4. Geometrical designation of roll groove and the radius of surface profile, R_S, of a workpiece in oval–round pass rolling.

groove is not a round shape any more. It rather closes to an oval shape. Since the maximum spread formula [6] for oval–round pass rolling was developed on the basis of perfect round groove rolling, only perfect round groove was considered in this study.

The cross points, C_x and C_y (Fig. 4), can be obtained by solving two simultaneous circular equations for the round roll groove and the surface profile of workpiece. Then, the cross sectional area of workpiece in round pass is expressed by Eq. (6)

$$A_{\text{round}} = 4\int_0^{C_x} (R_g{}^2 - x^2)^{1/2}\,\mathrm{d}x - R_S{}^2(2\theta - 2\sin 2\theta) \qquad (6)$$

where

$$\theta = \sin^{-1}\left(\frac{C_y}{R_S}\right)$$

2.3. Surface profile and cross sectional area of workpiece in round–oval pass rolling

Fig. 5(a) describes a round–oval pass rolling. R_a is the radius of curvature of an incoming round workpiece and R_1 is the radius of curvature of oval groove. Since it is inherently impossible to define the radius of a final surface profile in oval groove, the linear interpolation technique employed in oval–round pass rolling can not be used directly. Here, an assumption is introduced to get the radius of the final surface profile in oval groove. According to this assumption, the incoming round workpiece spreads to the roll axis (gap) direction and at last reaches at the face width of oval groove, W_f, when it is deformed at the inside of oval groove. Then, as shown in Fig. 5(b), the final surface profile is assumed to be a circle with a radius of R_f. R_f is centered at some point of x-co-ordinate. Henceforth f R_f is obtained, the linear interpolation technique employed in oval–round pass rolling can be used in round–oval pass rolling immediately. This simplification is made to facilitate the solution of surface profile problems, which would otherwise be difficult to solve. Derivation procedure of the equations for R_f is described in Appendix A.

3. Experiment

3.1. Material and specimen preparation

Low carbon plain steel (0.1%C) was used in the experiments. The material was obtained in the form of hot-rolled squares with 160 mm. The specimens to be rolled were cut and machined into round section with 60 mm diameter and 300 mm length. The entry part of specimen was tapered out in order for the specimen to squeeze in the rolls easily. In order to measure the rolling temperature of workpiece, a thermocouple (type K) with 1.6 mm outside diameter was embedded in 40 mm deep holes drilled in the tail ends of the specimen.

3.2. Rolling equipment

A hot rod rolling experiment with a single stand mill was used, a two-high laboratory mill driven by a 75 kW constant torque, dc motor, Ductile Casting Iron (DCI) rolls were used, with 310 mm maximum diameter and 320 mm face width. The surface finish of the rolls was obtained by sand blasting. Rolling speed was set at 34 rpm. A box type furnace with the working temperature of 1400°C was employed to heat up the specimens to the desired rolling temperature.

The roll has one round groove and two oval grooves. The dimensions of grooves and roll gaps are illustrated in Fig. 6. At each pass, a static guide was installed in front of the roll groove.

3.3. Experimental procedure

The initial round specimen was heated up to 1030°C in an atmosphere of nitrogen and rolled at 1000°C without lubrication. Round specimen is first rolled into the oval pass in Fig. 6(a) or (b). The workpiece produced at oval pass is rotated 90° along the length direction and rolled into the round pass in Fig. 6(c).

The workpiece after each rolling pass was cooled in air. Cross-sections with 30 mm thickness were obtained by cutting the middle part of the long workpiece. Then, milling machine smoothed the cross section of the workpiece. Finally, the co-ordinates of surface profile were obtained

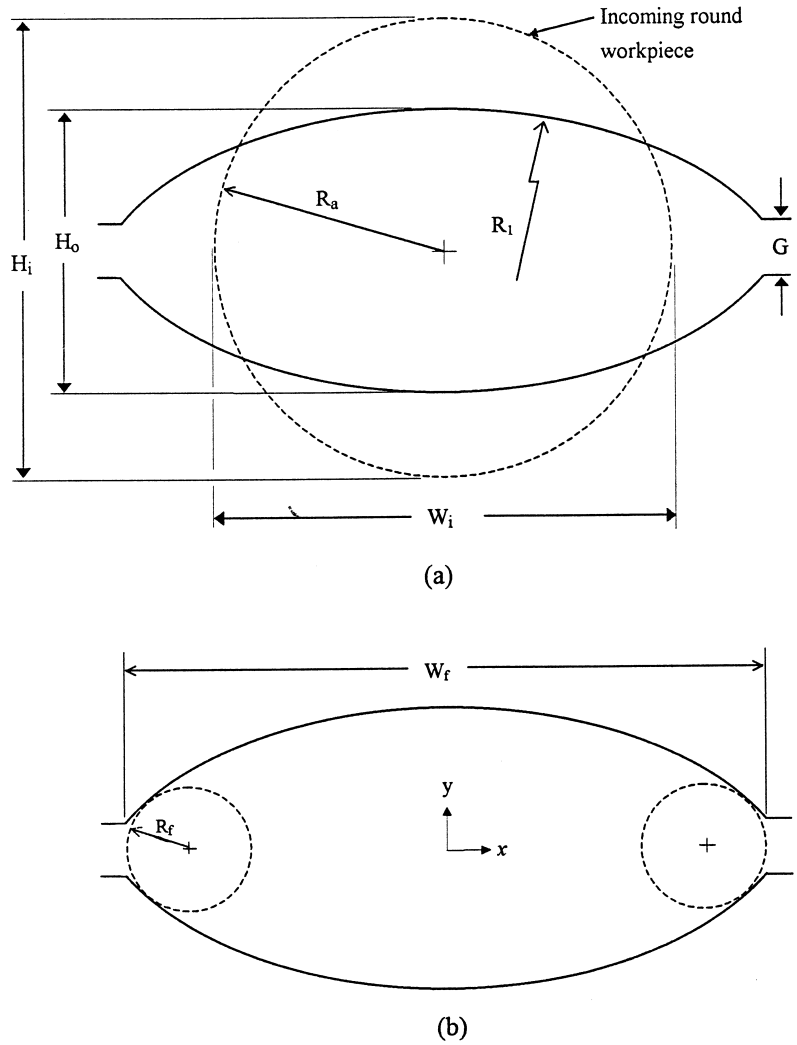


Fig. 5. (a) A round workpiece incoming and an oval groove, (b) Concept of the radius of the surface profile, R_f, in oval pass rolling.

by using a surface profile reading program [7] followed by scanning of the cross section of the workpieces.

4. Results and discussion

In Figs. 7–9, the surface profiles of workpiece obtained by experiment are compared with those predicted by the analytical model proposed. Dotted lines represent the roll groove shape and solid lines the predicted surface profile of the workpieces. The measured co-ordinate is shown by symbols indicating every third data point, for clarity.

The cross sectional area measured at room temperature after experiment is inevitably smaller than that predicted one because the surface layers oxidized by air were scaled off after air cooling of the hot workpiece.

4.1. Oval (B)–round pass rolling

Fig. 7 shows the predicted and measured surface profile of workpiece when the reference roll gap (6.5 mm) is used. The predicted radius of surface profile of workpiece is slightly larger than the measured one. In overall, however, the predicted surface profile agrees with the measured one. Fig. 8(a) illustrates that the experimental surface profile is in agreement with the predicted one when the roll gap increases to 10.5 mm. On the other hand, the predicted maximum spread is larger than the measured one when the roll gap decreases to 2.5 mm (Fig. 9(a)). However, this big difference could be partly caused by mistake in roll gap setting. Careful comparison of oval groove profile with experimentally measured profile of deformed workpiece shows that the roll gap in oval (B) pass was slightly larger than 2.5 mm during experiment. Hence, the height of workpiece became larger than the pass height of oval groove. Consequently, the measured maximum spread could be smaller than the predicted one.

In Fig. 8(b), the measured surface profile does not look like a round shape compared with the predicted one, which is almost a round shape. The workpiece in round groove has been rotated somehow along its length direction. This result

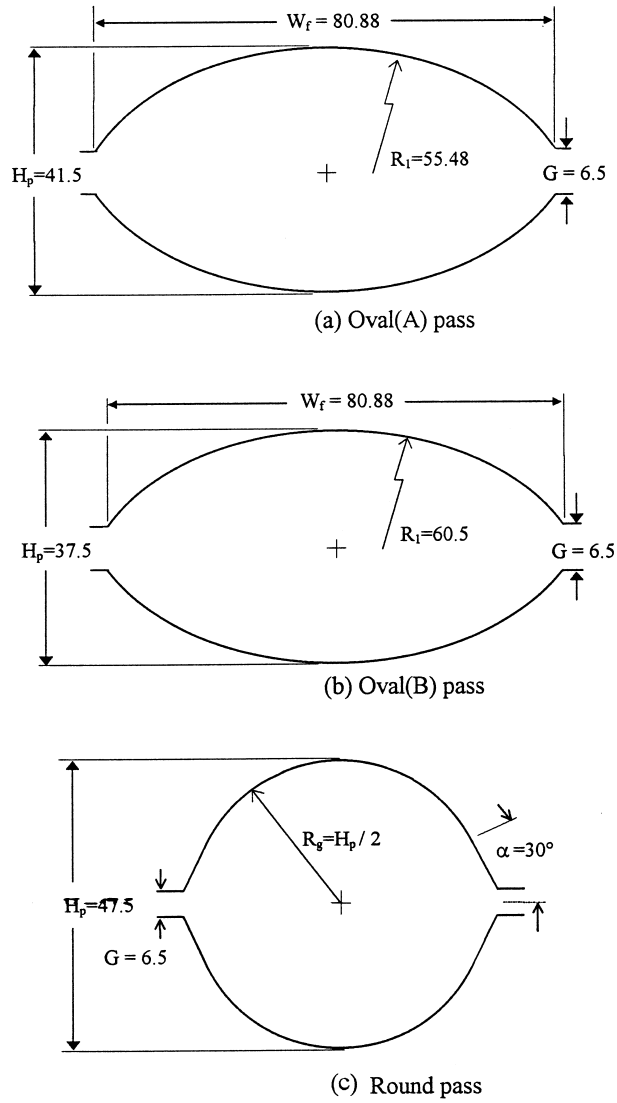


Fig. 6. Oval–round pass sequence designed for the hot rod rolling experiment. (a) Oval (A) pass, (b) Oval (B) pass, (c) Round pass.

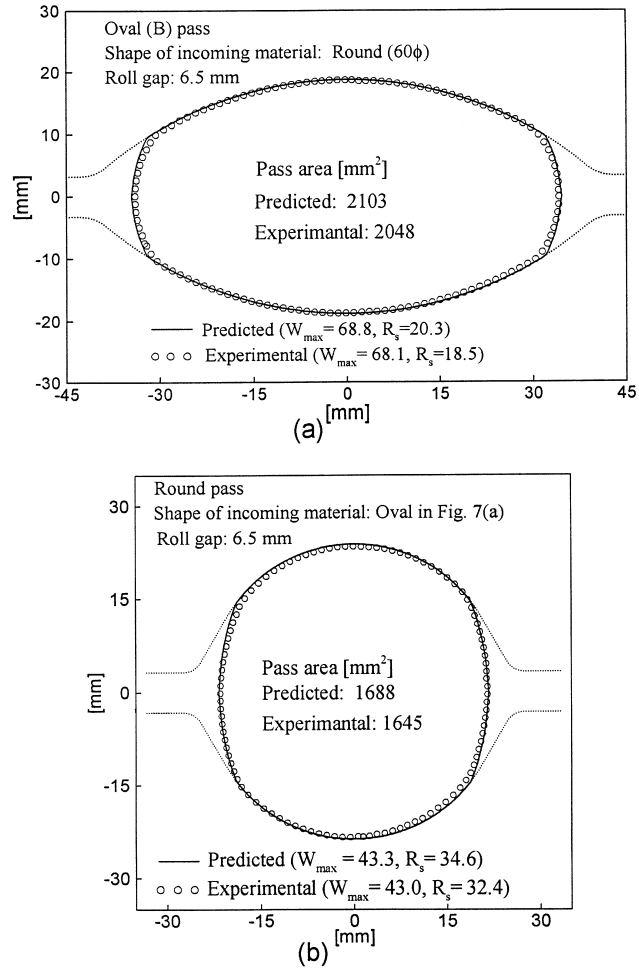


Fig. 7. Predicted and measured surface profile of workpiece in (a) oval (B), (b) round pass rolling (roll gap of oval groove = 6.5 mm).

seems to come from machine error such as loose clearance between the entry guide and workpiece. Fig. 9(b) shows that the maximum spread calculated by the analytical model is larger than that determined by experiment. This is quite natural because the maximum spread of the workpiece predicted by the analytical model (Fig. 9(a)) was already larger than that by experiment in the previous oval (B) pass.

4.2. Radius of surface profiles and cross-sectional area as a function of roll gap

For oval (A)–round pass rolling, the predicted and measured surface profile of workpiece were not presented. Instead, the radii of surface profile, R_S, and cross sectional area of workpiece experimentally measured and predicted by the model for oval (A)–round and oval (B)–round pass rolling are summarised in Figs. 10 and 11. Oval (A) groove has smaller radius of curvature, R_1, than oval (B) groove (see Fig. 6).

In Fig. 10(a) and (b), the experimentally measured radius of surface profiles are compared with the predicted one, with the roll gap of oval grooves changed. The predicted radii of surface profile in oval (A) pass are always larger (about 2.5 mm) than those obtained experimentally. In oval (B) pass, the differences between the radii of surface profile predicted and that measured become small as the roll gap decreases. But, the differences start to increase as roll gap of oval (B) groove increases.

Normalised cross-sectional area (A_{pred}/A_{expt}) of workpiece with roll gap changed in oval–round pass rolling sequence is illustrated in Fig. 11(a) and (b). In Fig. 11(a), a data for the oval (A)–round pass rolling with roll gap of 10.5 mm was missing because the test was invalidated by overfilling in round pass. Result from both oval (A)–round pass and oval (B)–round pass rolling shows that the differences of measured cross sectional area and predicted one are in the range of 1.5–3.8% when the roll gap of oval pass is changed up to ±61.5%.

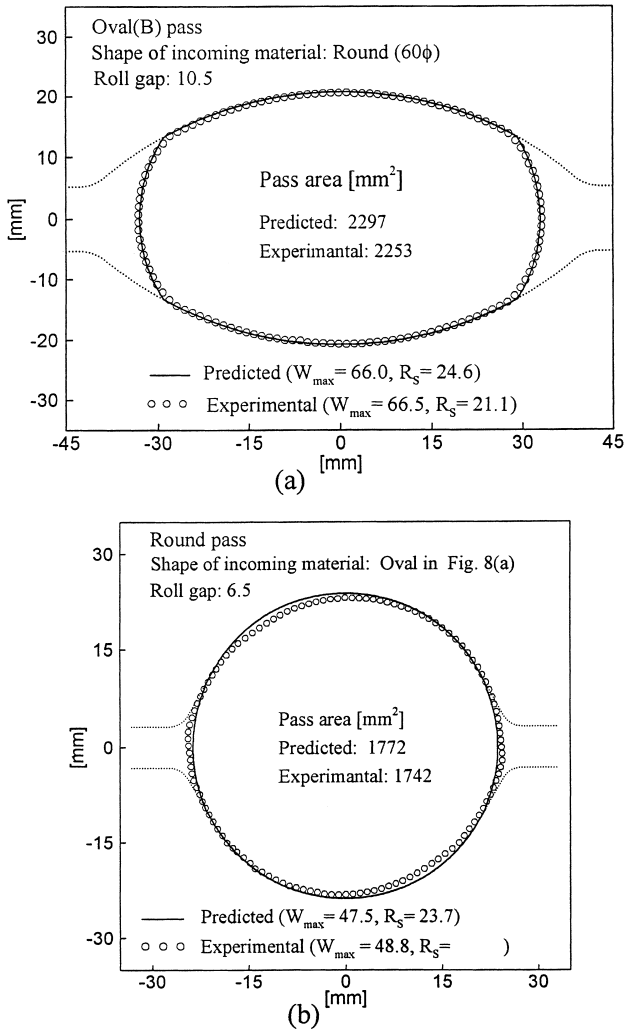


Fig. 8. Predicted and measured surface profile of workpiece in (a) oval (B), (b) round pass rolling (roll gap of oval groove = 10.5 mm).

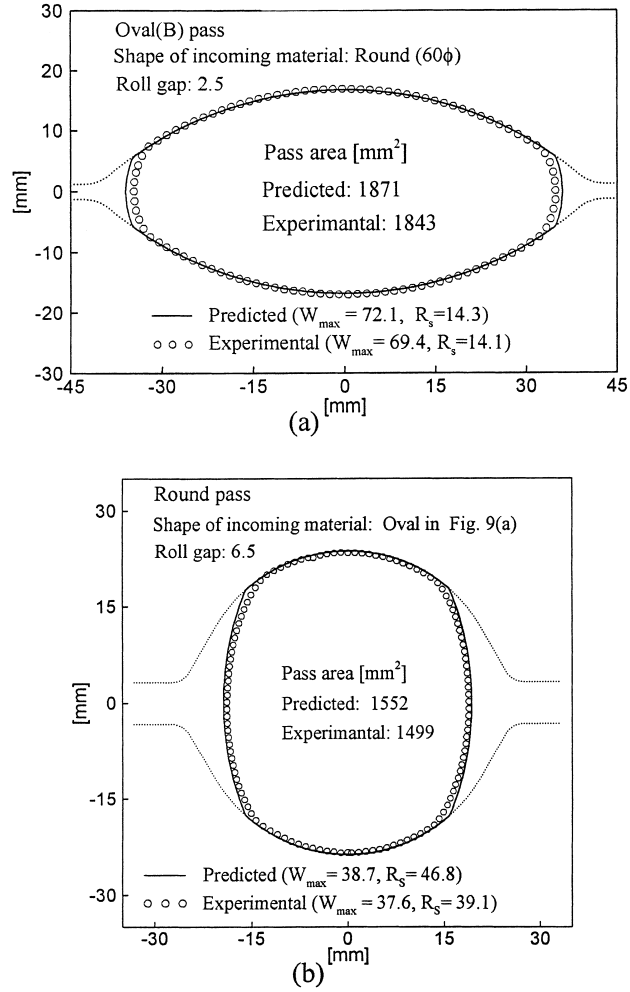


Fig. 9. Predicted and measured surface profile of workpiece in (a) oval (B), (b) round pass rolling (roll gap of oval groove = 2.5 mm).

5. Conclusions

In this study, an analytical model for the prediction of the surface profile of a workpiece during round–oval–round pass sequence has been proposed, and validity of the model has been examined by hot rod rolling experiment, with the roll gap and the design of oval groove changed. The conclusions are summarised as follows:

1. The surface profile of a workpiece predicted by the proposed analytical model is in agreement with those obtained experimentally for limited rolling conditions (groove shape and roll gap change).
2. The proposed model is powerful in predicting cross sectional area of workpiece from the point of view of computational time, compared with finite element simulation. But further studies are necessary to insure the model for more cases, including the change of rolling speed, stock geometry and the material.
3. The proposed model might potentially be a valuable tool for the on-line control of a round–oval–round pass sequence in continuous rod (or bar) rolling process.

Appendix A.

In Fig. 12, the equation describing the line O_1–B in x_1–y co-ordinate system is expressed as

$$y = ax_1 \tag{A.1}$$

where

$$a = \frac{R_1 - H_p/2}{W_f/2 - R_f} \tag{A.2}$$

An equation representing the upper part of oval groove can be expressed as

$$x_1^2 + y^2 = R_1^2 \tag{A.3}$$

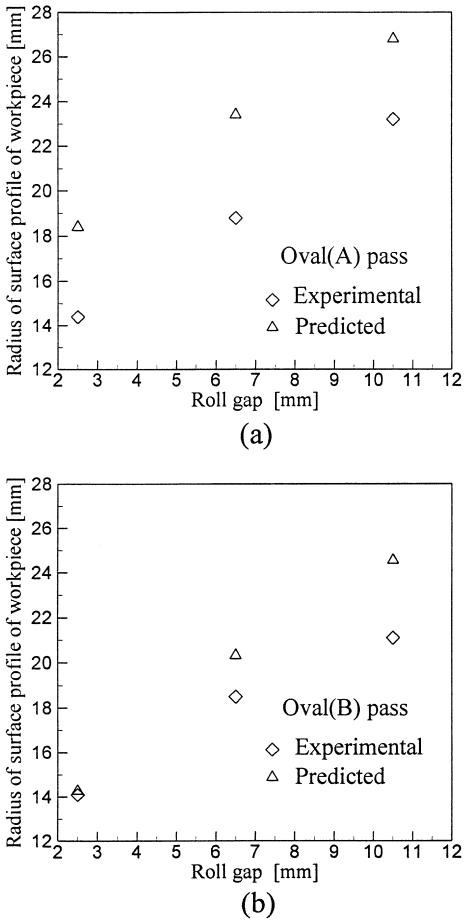


Fig. 10. Predicted and measured radius of surface profile of workpiece with the roll gap changed. (a) Oval (A) pass, (b) Oval (B) pass.

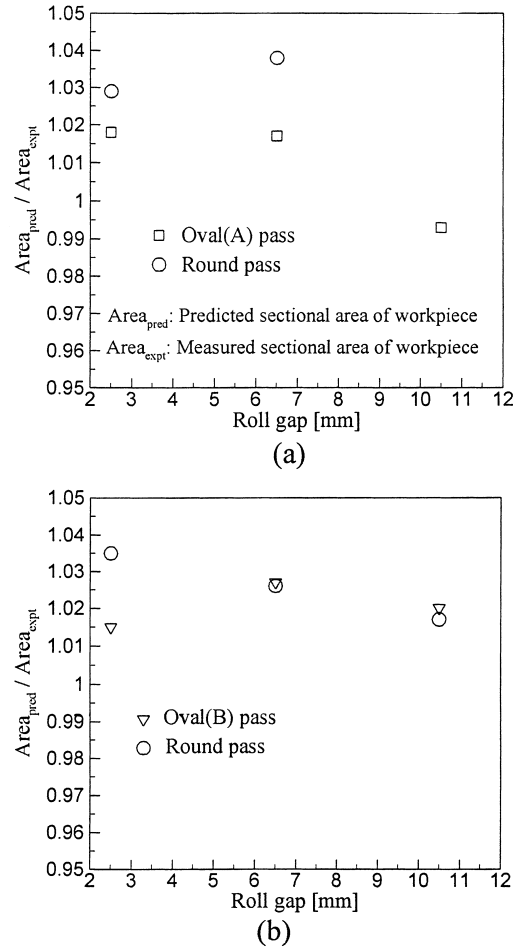


Fig. 11. Normalised cross sectional area with the roll gap changed. (a) Oval (A)–round pass sequence, (b) Oval (B)–round pass sequence.

Substituting Eq. (A.1) into (A.3) yields y-co-ordinate of point B with origin of O_1

$$B_y = \left(\frac{{R_1}^2 a^2}{1+a^2}\right)^{1/2} \tag{A.4}$$

The co-ordinate, B_y, can also be obtained from the geometric similarities

$$B_y = h + \left(\frac{R_1 - H_p}{2}\right) \tag{A.5}$$

where

$$h = \frac{R_f(R_1 - H_p/2)}{R_1 - R_f}$$

The term 'h' in Eq. (A.5) is derived from the geometric similarities of two triangles, $\Delta O_1 O O_2$ and $\Delta B O_3 O_2$, i.e.

$$\frac{h}{R_f} = \frac{(R_1 - H_p/2)}{R_1 - R_f} \tag{A.6}$$

Then, Eq. (A.5) can be rewritten

$$B_y = (R_1 - H_p/2)\left(\frac{R_1}{R_1 - R_f}\right) \tag{A.7}$$

Equating Eqs. (A.4) and (A.7) gives

$$\left[\frac{(R_1 - H_P/2)^2}{a^2}\right]\left[\frac{{R_1}^2}{(R_1 - R_f)^2}\right] = \frac{R_1^2}{1+a^2} \tag{A.8}$$

Arrangement by using Eq. (A.2) gives

$$\frac{(W_f/2 - R_f)^2}{(R_1 - R_f)^2} = \frac{1}{1 + ((R_1 - H_p/2)/(W_f/2 - R_f))^2} \tag{A.9}$$

Rearranging Eq. (A.9) yields

$$\left(\frac{W_f}{2 - R_f}\right)^2 + \left(\frac{R_1 - H_p}{2}\right)^2 = (R_1 - R_f)^2 \tag{A.10}$$

Thus, R_f can be expressed in terms of R_1, H_p and W_f

$$R_f = \frac{R_1 H_p - (1/4)({W_f}^2 + {H_p}^2)}{2R_1 - W_f} \tag{A.11}$$

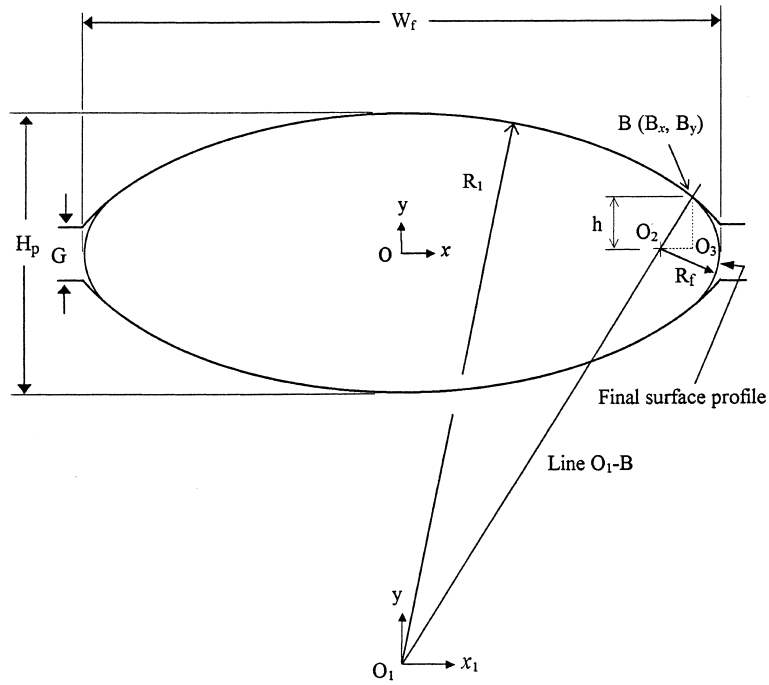


Fig. 12. The radius of final surface profile (R_f) of a workpiece in oval pass rolling.

Once R_f is obtained, R_S can be formulated by the linear interpolation of R_a and R_f, i.e.

$$R_S = R_a W_t + R_f(1 - W_t) \tag{A.12}$$

where

$$W_t = \frac{W_f - W_{max}}{W_f - W_i} \tag{A.13}$$

W_t is a weighing function of R_a and R_f, and W_f is the face width of oval groove. Hence, in round–oval pass rolling, one of the possible surface profiles, R_S can be illustrated as shown in Fig. 13. When R_S is calculated from Eq. (A.12), the distance between the center of oval groove and that of surface profile is computed

$$D_x = \frac{W_{max}}{2 - R_S} \tag{A.14}$$

The position of D_x is always along the x-axis.

In order to check the validity of Eq. (A.12), boundary conditions of the proposed model are examined. If the

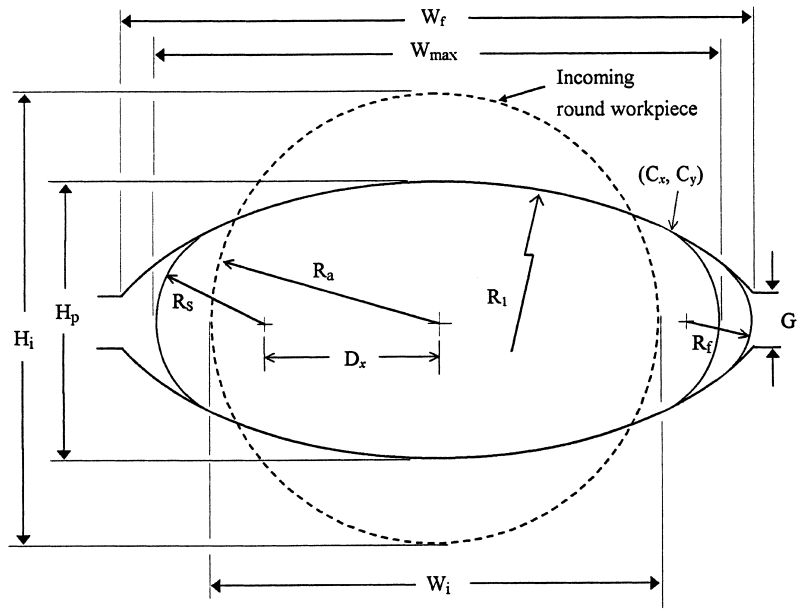


Fig. 13. Geometrical designation of roll groove and the radius of surface profile (R_f) of a workpiece in round–oval pass rolling.

incoming round workpiece with the radius of R_a is not deformed, there is no spread at all. Then, W_{max} (Fig. 13) is just W_i and $W_t = 1.0$ is obtained from Eq. (A.13). Consequently, Eq. (A.12) yields that $R_S = R_a$, which makes sense. On the other hand, if the maximum spread reaches the face width of oval groove after rolling, i.e., $W_{max} = W_f$, W_t in Eq. (A.13) is equal to zero. This implies that R_S in Eq. (A.12) becomes R_f, which is correct. It should be noted that Eqs. (A.11)-(A.13) are valid under the condition that the maximum spread of the outgoing workpiece is not greater than the face width of oval groove.

The cross points, C_x and C_y (Fig. 13), are obtained by solving two simultaneous circular equations for oval roll groove and the surface profile of workpiece. Then, the cross sectional area of workpiece in oval pass is computed by Eq. (A.15)

$$A_{oval} = 4\left[\int_0^{C_x} ({R_1}^2 - x^2)^{1/2}\,dx - \left(\frac{R_1 - H_p}{2}\right)C_x + \frac{\pi}{4}{R_S}^2 - \int_0^{W_{max}/2 - C_x} ({R_S}^2 - x^2)^{1/2}\,dx\right] \quad \text{(A.15)}$$

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